

```
## Claude Code v2.1.284
## Sonnet 5.5 · Claude Pro
## D:\SEM5\SE\lab3\28_defender
```

Opus 5.5 is now your default model and it draws down usage faster than Sonnet 5. Switch anytime with /model.

> The radar strip is supposed to show every humanoid, lander, and the player at their true position across the entire wrapping world, so threats anywhere in the world are visible at a glance. In the current build, the radar only ever shows things clustered near the player's own position, because it plots each blip's position relative to the player (screen space) instead of its absolute position in the world. Fix draw_radar so each blip's horizontal position on the radar reflects its actual place in the world, from one edge to the other, regardless of where the player currently is.

Searched for 2 patterns, read 1 file (ctrl+o to expand)

- Update(game.py)
 - | Added 1 line, removed 1 line
 - 203 blips += [(l.x, l.y, (255, 90, 90) if l.mutant else (255, 200, 60)) for l in self.landings]
 - 204 blips.append((self.player.x, self.player.y, (255, 255, 255)))
 - 205 for x, y, color in blips:
 - 206 - rx = self.screen_x(x) % VIEW_W
 - 206 + rx = (x % WORLD_W) / WORLD_W * VIEW_W
 - 207 ry = (y - PLAY_TOP) / (VIEW_H - PLAY_TOP) * (RADAR_H - 8) + 4
 - 208 pygame.draw.rect(screen, color, (rx - 2, ry - 2, 4, 4))
 - 209
- I fixed draw_radar in game.py. Each blip's horizontal position is now its absolute world position scaled across the radar: $rx = (x \% WORLD_W) / WORLD_W * VIEW_W$. Humanoids, landers and the player therefore spread from the left edge to the right edge of the radar according to where they are in the world, wherever the player is. I haven't run the game to check it visually.

The old code used `self.screen_x(x)`, which is the offset from the player's position, so the radar only showed things near the player.

* Cogitated for 9s · done 12:09 PM

> Called once per frame in draw, as `screen.fill(sky_color(self.wave) or (5, 5, 20))`. It receives the current wave number and should return an (r, g, b) color, or None to keep the default near-black sky. Idea: brighten or shift hue as waves progress.

Searched for 1 pattern (ctrl+o to expand)

- Update(game.py)
 - | Error editing file
- Bash(python -c "
import game...
| pygame 2.6.1 (SDL 2.28.4, Python 3.10.11)
Hello from the pygame community. <https://www.pygame.org/contribute.htm>
1
... +6 lines (ctrl+o to expand)
- Update(game.py)
 - | Added 9 lines, removed 1 line
 - 13
 - 14 def sky_color(wave):
 - 15 """Return an (r, g, b) sky colour for the current wave, or None for the default."""
 - 16 - pass
 - 16 + if wave <= 1:
 - 17 + return None
 - 18 + # Ease brightness up with the wave (capped so sprites stay +readable) and
 - 19 + # drift each channel on its own phase so the hue shifts wave to wave.
 - 20 + lift = min((wave - 1) * 6, 60)
 - 21 + return tuple(
22 + int(min(255, max(0, base + lift * 0.5 + 35 * math.sin(wave * 0.6 + phase))))
23 + for base, phase in zip((5, 5, 20), PHASES)
24 +)
25
26
27 def on_humanoid_rescued(humanoid):

- Bash(`python -c "`
`import game...`
`| 1 None`
`2 (10, 0, 7)`
`5 (50, 29, 66)`
`... +3 lines (ctrl+o to expand)`
- I implemented `sky_color` in `game.py`. I ran it for a few wave numbers and it returns valid colors, but I haven't looked at it in the running game.
 - Wave 1: it returns `None`, so you get the default near-black sky.
 - Later waves: it returns an `(r, g, b)` tuple. The base color brightens by about 3 per channel for each wave, up to a cap of 30, so the yellow and red sprites and the white player stay readable.
 - Hue shift: each channel also drifts on its own sine wave, using the `PHASES` list that was already defined at the top of the file. The hue therefore changes from wave to wave and is different each run.
 - Clamping: values are clamped to 0-255.

Sample output:

Wave	Sky color
2	(10, 0, 7)
5	(50, 29, 66)
10	(0, 24, 14)
50	(15, 62, 47)

The sine swing is ± 35 , larger than the base color, so a channel can hit 0 (wave 10 has red at 0). The sky stays dark everywhere, but shrink the 35 if you want it subtler.

* Churned for 19s · done 12:11 PM

- > Called from `update()` the moment the player catches a falling humanoid (one a lander dropped or released mid-air), right after the 500-point bonus is added and the humanoid is placed safely back on the ground. It receives the rescued Humanoid. Its return value is ignored. Idea: a "+500" popup, or a short invulnerability boost as a reward.

Searched for 1 pattern, read 1 file (ctrl+o to expand)

- Since the hook only receives the humanoid, I'll have it queue a popup in a module-level list that Game ticks and draws.
- Update(`game.py`)
 - | Added 1 line, removed 1 line


```
26
27 def on_humanoid_rescued(humanoid):
28     """Called when the player catches a falling humanoid; add a
        bonus or celebration here."""
29 -     pass
29 +     RESCUE_POPUPS.append({"x": humanoid.x, "y": humanoid.y - 20
        +, "life": POPUP_LIFE})
30
31
32 def bonus_life_threshold():
```
- Update(`game.py`)
 - | Added 2 lines


```
9 HUMANOID_COUNT = 6
10 FALL_LIMIT = 160
11 PHASES = [random.uniform(0, math.tau) for _ in range(3)]
12 +POPUP_LIFE = 1.0
13 +RESCUE_POPUPS = [] # floating "+500" labels: {"x": world x, "y
        +": screen y, "life": seconds left}
14
15
16 def sky_color(wave):
```
- Update(`game.py`)
 - | Added 4 lines


```
157         return
158         player = self.player
159         player.update(dt, keys)
160 +         for popup in RESCUE_POPUPS:
```

```

161 +         popup["life"] -= dt
162 +         popup["y"] -= 40 * dt
163 +         RESCUE_POPUPS[:] = [p for p in RESCUE_POPUPS if p["lif
+e"] > 0]
164 +         threshold = bonus_life_threshold()
165 +         if threshold and self.score // threshold > self.bonus_
awarded:
166 +             self.bonus_awarded = self.score // threshold

```

- Update(game.py)

```

| Added 4 lines
242 +         if player.invulnerable <= 0 or int(player.invulnerable
* 10) % 2 == 0:
243 +             f, cx = player.facing, VIEW_W / 2
244 +             pygame.draw.polygon(screen, (240, 240, 250), [(cx
+ f * 18, player.y), (cx - f * 14, player.y - 8), (cx - f * 14
, player.y + 8)])
245 +             for popup in RESCUE_POPUPS:
246 +                 label = self.font.render("+500", True, (120, 255,
+150))
247 +                 label.set_alpha(int(255 * popup["life"] / POPUP_LI
+FE))
248 +                 screen.blit(label, label.get_rect(center=(self.scr
+een_x(popup["x"]), popup["y"])))
249 +                 self.draw_radar(screen)
250 +                 hud = self.font.render(f"Score {self.score} Lives {se
lf.lives} Wave {self.wave} Humanoids {len(self.humanoids)}",
True, (240, 240, 240))
251 +                 screen.blit(hud, (10, RADAR_H + 4))

```

- Update(game.py)

```

| Added 1 line
137 +         self.landings, self.bullets = [], []
138 +         self.score, self.lives, self.wave, self.state = 0, 3,
1, "play"
139 +         self.bonus_awarded = 0
140 +         RESCUE_POPUPS.clear()
141 +         self.start_wave()
142 +
143 +         def start_wave(self):

```

- Bash(SDL_VIDEODRIVER=dummy python -c "

```

import pygame,game...)
| Hello from the pygame community. https://www.pygame.org/contribute.htm
1
1
0

```

- I implemented on_humanoid_rescued in game.py as a floating green "+500" popup.

I checked it headlessly: the popup is created, drawn and expires after about a second. I haven't looked at it in the actual game window.

- Hook: it adds a popup at the rescued humanoid's position, slightly above it. It only receives the humanoid, so the popup is kept in a new module-level list, RESCUE_POPUPS.
- Game.update: each frame the popups drift upward and their remaining time counts down. Expired ones are removed.
- Game.draw: each popup is drawn at its world position through screen_x, so it stays with the humanoid as the camera moves. It fades out over POPUP_LIFE (1 second).
- Game.reset: it clears the list, so no popups carry over into a new game.

I didn't add the invulnerability boost. The hook has no access to the player, so that would need either a global reference or a change to the call site in update(). Say if you want it.

* Crunched for 23s · done 12:12 PM

> Called every frame in update(). It takes no arguments and should return an integer score value, or None to disable bonus lives entirely. Whenever the score crosses a multiple of that value for the first time, one life is awarded automatically – the bookkeeping (self.bonus_awarded) is already implemented, so you only need to choose the threshold. Idea: return 10000.

- Update(game.py)

```

| Added 1 line, removed 1 line
33
34 + def bonus_life_threshold():
35 +     """Return a score value at which the player earns an extra
life, or None to disable bonus lives."""
36 -     pass
36 +     return 10000

```

```
37
38
39 def wrap_delta(a, b):
```

- bonus_life_threshold() in game.py now returns 10000, so the player gets an extra life each time the score first crosses a multiple of 10,000. I didn't run the game to test it.

* Crunched for 4s · done 12:12 PM

```
> commit these changes without coauthoring also follow
@Lab_4_VibeCoding_handout.pdf and give all the handouts'
| Read Lab_4_VibeCoding_handout.pdf (69.7KB)
```

Read 1 file (ctrl+o to expand)

- Bash(python -c "
import pypdf...)
| Traceback (most recent call last):
File "<string>", line 4, in <module>
File "C:\Users\Lenovo\AppData\Local\Programs\Python\Python310\lib\en
... +4 lines (ctrl+o to expand)

- Bash(PYTHONIOENCODING=utf-8 python -c "
import pypdf...)
| Lab 4: VibeCoding
Objective
Use Vibe Coding tools to fix the broken code and add featur
... +73 lines (ctrl+o to expand)

- Bash(git status --short && ls -la && git remote -v && git diff --stat)
| M game.py
?? Lab_4_VibeCoding_handout.pdf
total 101
... +13 lines (ctrl+o to expand)

Searched for 1 pattern (ctrl+o to expand)

- Checking the handout against the repo: it wants a Lab-4 folder with code, videos and chat history. I'll commit game.py and the handout PDF into that structure, with no co-author line.

- Bash(mkdir -p Lab-4 && cp game.py Lab-4/game.py && git mv -f
Lab_4_VibeCoding_handout.pdf Lab-4/ 2>/dev/null || mv
Lab_4_VibeCoding_handout.pdf Lab-4/ && git add gam...)
| 42827ce Fix radar, add sky colour, rescue popup and bonus life thresho
ld
... +5 lines (ctrl+o to expand)

- I committed the changes as 42827ce with no co-author line. I haven't pushed it.

Handout summary (Lab 4: VibeCoding). You fix the broken code and add the README features using an LLM, then push to your own repo, not a PR to the main repo. The individual assignment is 60 minutes long. The deliverables go in a Lab-4 folder in your repo:

1. Before and after videos, 10 seconds each.
2. The updated code.
3. The chat history, exported as a doc or PDF.

The README's submission checklist matches this, except it asks for a chat link instead of an export.

What's in the commit:

- game.py with the radar fix, sky_color, on_humanoid_rescued and the 10000 bonus life threshold.
- Lab-4/game.py, a copy of the updated code.
- Lab-4/Lab_4_VibeCoding_handout.pdf, the handout.

Still missing, and I can't produce these:

- The 10-second before video, which had to be recorded before any changes.
- The 10-second after video.
- The chat history export or link.

Add those to Lab-4/, then push with git push. Say if you'd like me to push once they're in.

* Cogitated for 25s · done 12:13 PM

> push it

- Bash(git push origin main 2>&1 | tail -5)
| To https://github.com/DhruvaMyakeri/28_defender

031f1ee..42827ce main -> main

- Pushed commit 42827ce to origin/main at github.com/DhruvaMyakeri/28_defender. You still need to add the before and after videos and the chat history to Lab-4/.

* Brewed for 9s · done 12:14 PM

* recap: Goal is completing the Lab 4 Defender submission: the radar fix, sky colour, rescue popup and bonus life are committed and pushed to main. Next, add the before and after videos and the chat history to the Lab-4 folder and push again. (disable recaps in /config)

> check

- Bash(`git fetch -q && git status -sb | head -5 && git log --oneline -2 && git ls-tree -r --name-only origin/main && diff game.py Lab-4/game.py && echo same && SDL_VID...`)
| ## main...origin/main
| ?? "Screen Recording 2026-09-30 120032.mp4"
| ?? "Screen Recording 2026-09-30 121840.mp4"
| ... +12 lines (ctrl+o to expand)
| Interrupted · What should Claude do instead?

> check I have uploaded the recordings

- Bash(`ls -l *.mp4 && git ls-tree -r --name-only origin/main | grep -i mp4 || echo "no mp4 on origin/main"`)
| -rw-r--r-- 1 Lenovo 197121 20458676 Sep 30 12:00 Screen Recording 2026-09-30 120032.mp4
| -rw-r--r-- 1 Lenovo 197121 6887211 Sep 30 12:18 Screen Recording 2026-09-30 121840.mp4
| ... +2 lines (ctrl+o to expand)
- The recordings are not uploaded yet. They're in the repo folder, but git hasn't picked them up and they aren't on origin/main.
 - Screen Recording 2026-09-30 120032.mp4 (about 20 MB), recorded at 12:00. This is presumably the before video.
 - Screen Recording 2026-09-30 121840.mp4 (about 7 MB), recorded at 12:18. This is presumably the after video.

They are also in the repo root rather than in Lab-4/, where the handout wants them. Both are under GitHub's 100 MB file limit.

I can move them into Lab-4/ as before.mp4 and after.mp4, commit and push. Should I? I haven't checked which one is which, so tell me if the order is different. The chat history is still the remaining missing item.

* Churned for 6s · done 12:21 PM

> ok, and check if we have all the deliverables

- Bash(`mv "Screen Recording 2026-09-30 120032.mp4" Lab-4/before.mp4 && mv "Screen Recording 2026-09-30 121840.mp4" Lab-4/after.mp4 && git add Lab-4 && git commit -q -m...`)
| To https://github.com/DhruvaMyakeri/28_defender
| 42827ce..f859544 main -> main
| Lab-4/Lab_4_VibeCoding_handout.pdf
| ... +67 lines (ctrl+o to expand)
- I moved the two recordings into Lab-4/ as before.mp4 and after.mp4, committed them and pushed. origin/main is now at f859544.

Deliverables against the handout and README:

Deliverable	Status
Updated code	Done. Lab-4/game.py and root game.py are identical.
Before video	Done. Lab-4/before.mp4.
After video	Done. Lab-4/after.mp4.
Chat history (PDF, doc or link)	Missing.
README tasks 1-4 (radar, sky_color, rescue hook, bonus life)	All implemented.

- Chat history: the README wants the LLM chat link with the full history, and the handout wants it exported as a doc or PDF in Lab-4/. I can't export this

conversation. You need to share the chat link or export it yourself. If you drop the file in, I'll commit and push it.

- Video order: I assumed the 12:00 recording is the before and the 12:18 one is the after, based on timestamps. Check that before.mp4 really shows the broken radar.
- Testing: I ran the game headless for 120 frames without errors. I haven't played it, and nobody has confirmed the rescue popup or the bonus life in real play.

* Cogitated for 15s · done 12:21 PM

* recap: Goal: submit the Lab 4 Defender assignment. The code, before/after videos and handout are pushed to your repo under Lab-4; only the chat history is missing. Next, share the chat link or export it as a PDF into Lab-4 and I'll commit and push it.
· (disable recaps in /config)

> /export

| Export cancelled